

GAL-2 Node RC11 Linux AMD64 Security and Deployment Handoff

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September 5, 2026

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Document version: 1.0

Document date: 2026-09-05

Release: v1.0.0-rc11-amd64-evaluator

Audience: independent technical and security evaluators

1. Purpose and scope

This document provides the release identity, provenance boundary, security status, network behavior, host modifications, deployment sequence, fault injection behavior, failure criteria, and cleanup procedure for the GAL-2 Node RC11 Linux AMD64 evaluator.

This is a controlled technical evaluation candidate. It is not a declaration of general production readiness, independent certification, native AMD64 performance qualification, or a completed third-party security review.

The intended product boundary is:

reference inputs -> Protected Core -> GAL-2 Time -> API -> Node -> Time Contract -> enrolled application

GAL-2 does not replace or discipline the host clock, NTP, PTP, GNSS, chrony, or systemd-timesyncd. An enrolled application consumes GAL-2 Time through the GAL-2 Provider. The Time Contract field `safe_to_consume` is the authoritative application-consumption decision.

2. Direct answers to evaluator due-diligence questions

Question	Answer for this release
Is there an official download?	Yes. The frozen evaluator is distributed through the GitHub release identified below.
Is the download cryptographically identified?	Yes. The outer ZIP, signed package, detached signature, public key, evidence archive, runtime archive, OCI manifest, OCI config, and tested payload are identified by SHA-256.
Is the package signed?	Yes. The signed package has a detached OpenPGP signature. The expected primary fingerprint is published below.
Does a good signature prove trust in the supplied key?	No. It proves cryptographic validity against that key. The evaluator must verify the primary fingerprint through an independent trusted GAL-2 channel.
Is the distributed runtime built solely from source in the public evaluator repository?	No. The public repository does not contain the GAL-2 Protected Core, the complete internal runtime source, or a Dockerfile capable of rebuilding the frozen OCI image.
What does <code>Reproducible build: PASS</code> mean?	It applies to deterministic evaluator-package assembly over frozen inputs. It does not mean the OCI runtime can be independently rebuilt from the public repository.
Is an SBOM included?	No formal SPDX, CycloneDX, or equivalent SBOM is included in RC11.
Is a formal vulnerability scan report included?	No. RC11 does not include a formal CVE or container-vulnerability scan report. Absence of a report must not be interpreted as absence of vulnerabilities.
Was there an independent security audit?	No. The bundled audit and external-evaluation evidence are founder-operated. This handoff requests independent evaluation.
Does the package contain an API key?	No. A paid or temporary evaluator credential is supplied separately through a private channel.
What remote application destination is used?	Static inspection found one application-level remote destination: <code>https://api-v2.gal-2.com/time</code> .
Is telemetry sent to another destination?	No separate analytics, crash-reporting, or telemetry destination was found in the frozen GAL-2 application source. A local request counter is maintained, and the GAL-2 API necessarily observes authenticated API requests.
Does the Node alter the host timing stack?	No host-clock mutation primitive was found in the packaged deployment or evaluation scripts. The harness records timing-service state before and after the run and requires it to remain unchanged.
Is Podman supported?	No. Podman and <code>podman-docker</code> are not supported by this RC11 package. Genuine Docker Engine and <code>docker.service</code> are required.

3. Official release identity

Official release:

<https://github.com/gal-2-technologies/gal2-node-rc11-amd64-evaluator/releases/tag/v1.0.0-rc11-amd64-evaluator>

Required public downloads:

GAL2_NODE_LINUX_AMD64_RC11_EXTERNAL_AUDIT_FINAL_20260902.zip
 GAL2_NODE_LINUX_AMD64_RC11_EXTERNAL_AUDIT_FINAL_20260902.zip.sha256

GitHub-generated Source code (zip) and Source code (tar.gz) downloads are not the evaluator package.

3.1 Release artifacts

Identity	Value
Release tag	v1.0.0-rc11-amd64-evaluator
Package source revision	a22098a01462917ac4923ea9ffdb0a0f56e5c328
Final audit ZIP	GAL2_NODE_LINUX_AMD64_RC11_EXTERNAL_AUDIT_FINAL_20260902.zip
Final audit ZIP SHA-256	0db64868d6bfe1a7230a0e675fd24aec096e45a916e1813dcaac4d2d534efefb
Signed package	GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz
Signed package size	45323256 bytes
Signed package SHA-256	d452a276af1bf4bd2abb11dfa01f14f22f7c72f36eda9a3019c4269ef36ef553
Detached signature SHA-256	5c56f92df04f37d73b13625f06aafe2d1faf2d5f199c3007982f73aef2a48773
Release public key SHA-256	4342e36099b65889fc18fe06ef01063595ee15bd0d2d3df6cd3d81d4b28f1bfc
External evidence SHA-256	7777d141ef05245ff2cd76cba8eece9735b03474c134ecfb2f08bc5ee66d305f
Expected signing fingerprint	802C 8978 FF85 7550 60B6 D6BC 8AB8 59E4 D705 822F

The final audit ZIP contains exactly seven files:

GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz
 GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz.sha256
 GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz.asc
 GAL2_RELEASE_SIGNING_PUBLIC_KEY_RC11.asc
 AUDIT_IDENTITY.txt
 GAL2_NODE_LINUX_AMD64_RC11_EXTERNAL_EVIDENCE_A22098A_20260902.tar.gz
 GAL2_NODE_LINUX_AMD64_RC11_EXTERNAL_EVIDENCE_A22098A_20260902.tar.gz.sha256

3.2 Frozen runtime identity

Identity	Value
Runtime source revision	c8515d4feceda6eaa6fb3389f41a0b729493fc24
Runtime archive SHA-256	30361a33110d29b52c2d90ab5ddc0f0a479a60356a079fef3acdbbf87d901ad2
OCI manifest digest	sha256:497ab1646033ecbb28504489c874f6eda80b12507a7bd3ca4f2cb498fb471f9d
OCI config digest	sha256:abc2c606cba0e2b755398c51fd6919e5d8f2c880ebda11def76f6c8be567ad9f
OCI layers	11
Platform	linux/amd64
Original tested tag	gal2-node:rc11-amd64-evaluator-c8515d4
Time Contract	1.2.0-contract-rc.4
SHM layout	1.0
Frozen tested-payload manifest SHA-256	2ff1e8a8dfc3a80bce135727d921b760824c60d939c8321442d5b8788f444661
Tested-payload checksum-set SHA-256	f5ec86cfdc9334226f11c5738593c888974561695a26d6018c267a57be5bcd76

The identity policy declares:

```

consumer_payload_byte_for_byte_frozen=true
runtime_archive_byte_for_byte_frozen=true
runtime_rebuild_allowed_after_freeze=false
validation_evidence_is_runtime_payload=false

```

3.3 Critical deployment component identities

Component	SHA-256
Package builder	f48a84f6421630ddd596aa5ffac1883f98e81b4943280de5117055320309ae18
Installer	dc0e460f5172f127823081e2a9f37b78ef30c1cd24cecc7b80de7f17ab96b34e
Uninstaller	174f61680d79c9bc8abab549141939f9a378777b4a065551dcb2e613fbe14465
gal2-node control command	9e12d879e77a39a464becbf64e72dc87650876ec9cab55af385f716279dd5aab
Container guard	edad5fcdfa24eb53ea558a8068c8afce9faad000609500c1a7c7d4511d8cc4c0
Fault-injection firewall helper	83997a04b693cd71539870b0242825d3c2db684e3c8b4c93ebd9b0b5efc3db64
External evaluation harness	87824d26bf41d746d2c9b57fe62a3a216fdb4bf2a07ec095262ca4dc683ca7d
systemd unit template	bd87d969fdd9433e01424289edc4f38895f9a14aa9e41039bbb15b08e2049ebf

4. Independent download and signature verification

Run these commands from the download directory:

```

sha256sum --check \
  GAL2_NODE_LINUX_AMD64_RC11_EXTERNAL_AUDIT_FINAL_20260902.zip.sha256

mkdir gal2-rc11-amd64-evaluation
unzip GAL2_NODE_LINUX_AMD64_RC11_EXTERNAL_AUDIT_FINAL_20260902.zip \
  -d gal2-rc11-amd64-evaluation
cd gal2-rc11-amd64-evaluation

sha256sum --check \
  GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz.sha256
sha256sum --check \
  GAL2_NODE_LINUX_AMD64_RC11_EXTERNAL_EVIDENCE_A22098A_20260902.tar.gz.sha256

```

Verify the detached signature with an isolated temporary GPG home:

```

(
  set -e
  verify_home="$(mktemp -d)"
  trap 'rm -rf -- "$verify_home" EXIT'
  chmod 700 "$verify_home"

  gpg --batch \
    --homedir "$verify_home" \
    --import GAL2_RELEASE_SIGNING_PUBLIC_KEY_RC11.asc

  gpg --batch \
    --homedir "$verify_home" \
    --fingerprint

  gpg --batch \
    --homedir "$verify_home" \
    --verify \
    GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz.asc \
    GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz
)

```

Required results:

- GPG reports a good signature.
- The primary fingerprint exactly matches 802C 8978 FF85 7550 60B6 D6BC 8AB8 59E4 D705 822F.

- A warning that the key is not certified in the evaluator's temporary keyring is expected. Trust in the signing identity still requires an independent fingerprint comparison.

4.1 Independent fingerprint channel

For this evaluation, compare the fingerprint published in the GitHub handoff with the same fingerprint stated directly in a message from a GAL-2 corporate email address under `gal-2.com`. Confirm the sender through the LinkedIn account used to arrange the evaluation. The GitHub publication and the separately confirmed corporate message are distinct transport channels.

The copy of the fingerprint inside the downloaded ZIP is useful corroborating metadata, but it is not an independent channel because it travels with the artifact being verified.

Extract and verify the tested payload:

```
tar -xzf GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz
cd GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator
sha256sum --check SHA256SUMS.tested-payload.txt
```

Every listed file must report `ok`.

5. Build and source provenance boundary

The release separates package assembly identity from runtime build identity.

5.1 What is demonstrated

- The outer evaluator package has a declared package source revision.
- `PACKAGE_MANIFEST.json` declares a 21-file source allowlist plus generated package metadata.
- Packaged files have declared paths, sizes, modes, and SHA-256 values.
- `scripts/build_package.py` assembles the evaluator package over a previously frozen runtime archive and verifies the expected runtime archive SHA-256.
- The runtime OCI archive, manifest, config, layers, labels, platform, and tested consumer payload are cryptographically fixed.
- The installer refuses an unexpected archive, OCI graph, Docker object, platform, label set, file identity, or file mode.
- `AUDIT_IDENTITY.txt` records Reproducible build: PASS for this package assembly process.

5.2 What is not demonstrated publicly

- The public evaluator repository is not the complete source repository for the runtime.
- The GAL-2 Protected Core and backend are not included.
- The runtime Dockerfile and complete internal build recipe are not included.
- An evaluator cannot rebuild the OCI image solely from the public repository.
- The package and runtime source revisions are GAL-2 internal provenance declarations. They are not proof of equivalence to a complete public source tree.
- No third-party build attestation or independent reproducible-build result is included.

The accurate claim is therefore: the evaluator package is reproducibly assembled over frozen, cryptographically identified inputs. It is not a fully publicly reproducible build of the protected runtime.

6. Security assurance status

6.1 Included checks and evidence

- SHA-256 identity for the distribution and nested artifacts
- Detached OpenPGP signature and fixed expected primary fingerprint
- Package source allowlist and exact file modes
- Tested-payload checksum verification
- OCI graph and blob verification

- OCI platform and label verification
- Docker object identity verification after load
- Installed file ownership, mode, and hash validation
- Source secret scan recorded as `PASS`
- Evidence secret scan recorded as `PASS`
- Evidence-structure validation recorded as `PASS`
- Controlled functional and fault-injection evidence
- Refusal-oriented uninstall and purge ownership checks

6.2 Not included in RC11

- Formal SPDX or CycloneDX SBOM
- Formal container or dependency vulnerability scan report
- Independent penetration test
- Independent source-code security audit
- Independent supply-chain audit
- Publicly reproducible build of the protected OCI runtime
- Production certification

These absences are disclosed limitations. They must not be converted into claims that no vulnerabilities exist.

6.3 Visible Python distribution inventory

Static inspection of the exact frozen runtime found the following installed Python distributions. This is supplemental inventory, not an exhaustive SBOM of the operating-system packages, runtime, or build chain.

```
blinker==1.9.0
certifi==2026.7.22
charset-normalizer==3.4.9
click==8.4.2
Flask==3.1.3
idna==3.18
itsdangerous==2.2.0
Jinja2==3.1.6
MarkupSafe==3.0.3
pip==26.2.1
requests==2.34.2
urllib3==2.7.0
Werkzeug==3.1.8
```

7. Known static-analysis artifact

The frozen OCI runtime contains this generated bytecode file:

```
/opt/gal2/gate2/src/__pycache__/gal2_ixoye_observer.cpython-313.pyc
```

The corresponding source is included in the same frozen OCI runtime:

```
/opt/gal2/gate2/src/gal2_ixoye_observer.py
```

For a scanner that recognizes a URL across the intervening marshalled bytes, this exact `.pyc` deterministically produces the following concatenated finding. Expect it from permissive byte-oriented URL extraction:

```
http://127.0.0.1:9095/contractz(/usr/local/gal2/evidence/ixoye_live_rc57z
```

The source contains two separate values:

```
DEFAULT_CONTRACT_URL = "http://127.0.0.1:9095/contract"
DEFAULT_OUTPUT_DIR = "/usr/local/gal2/evidence/ixoye_live_rc57"
```

The actual observer network operation uses `urlopen` against the configured contract URL, whose default is the loopback endpoint. It does not use the concatenated string reported by byte-oriented scanners.

The file is timestamp-based bytecode. Its embedded source timestamp and source size match the included `gal2_ixoye_observer.py`. The image also sets `PYTHONDONTWRITEBYTECODE=1`; that setting prevents new bytecode writes but does not prohibit Python from loading valid existing bytecode.

The normal OCI command runs `/opt/gal2/gate2/src/gal2d.py`. Static inspection found no import or invocation of `gal2_ixoye_observer` from `gal2d.py`, so the normal Node entrypoint is not shown to load this `.pyc`. The bytecode remains potentially loadable if that observer module is explicitly imported or invoked, and it is therefore disclosed as part of the image's static-analysis and attack surface.

The `.pyc` is not individually listed as an outer package source file, but it is not outside the release identity. It is transitively frozen by its OCI layer digest, the OCI manifest digest, and the runtime archive SHA-256.

Generated-bytecode removal and an explicit build-time rejection invariant are tracked as RC12 build-hygiene and static-analysis improvements.

8. Runtime network behavior and telemetry

8.1 Host-facing interface

The process binds to port 9095 inside the container. Docker publishes it only on host loopback:

```
127.0.0.1:9095:9095
```

Principal local interfaces include:

```
http://127.0.0.1:9095/health
http://127.0.0.1:9095/contract
/run/gal2/contract-v1.shm
```

The service is not published on a non-loopback host interface.

8.2 Remote destination

Static inspection of the frozen GAL-2 application source found one actual application-level remote request:

```
GET https://api-v2.gal-2.com/time
x-api-key: <configured credential>
```

Default request behavior includes:

Setting	Default
Active refresh	30 seconds
Idle refresh	30 seconds
Holdover retry	10 seconds
HTTP timeout	10 seconds

Normal operation therefore requires DNS resolution and outbound HTTPS on TCP port 443 to `api-v2.gal-2.com`.

8.3 Non-destination URL strings

The runtime also contains these schema identifiers:

```
https://gal-2.com/schemas/gal2-time-contract-v1.schema.json
https://json-schema.org/draft/2020-12/schema
```

They identify JSON schemas. No application network call using either value was found in the inspected runtime source.

The IXOYE observer's default URL is local:

```
http://127.0.0.1:9095/contract
```

8.4 Telemetry boundary

- No separate remote analytics, advertising, crash-reporting, or telemetry destination was found in the inspected application source.
- Structured application logs are written to the container log stream and are available locally through `journalctl` and `gal2-node` logs.
- A local request count is persisted in `/var/lib/gal2/gal2_usage.json`.
- The configured GAL-2 API necessarily receives authenticated synchronization requests and can enforce the associated API entitlement and quota.
- The service loads the frozen image locally and uses `--pull never`; normal Node startup does not pull an image from a registry.

9. Host-side modifications

The installer requires root privileges and performs the following changes:

Host object	Purpose and expected state
<code>/etc/gal2/</code>	Dedicated configuration directory, root-owned, mode <code>0700</code>
<code>/etc/gal2/node.env</code>	API and Node configuration, root-owned, mode <code>0600</code> ; created with an empty API key if absent
<code>/var/lib/gal2/</code>	Persistent cache and local request state, root-owned, mode <code>0700</code>
<code>/run/gal2/</code>	Runtime directory and SHM publication, mode <code>0755</code>
<code>/run/gal2/contract-v1.shm</code>	Application-facing Time Contract SHM record, 8192 bytes, normally mode <code>0644</code>
<code>/usr/local/bin/gal2-node</code>	Control command, root-owned, mode <code>0755</code>
<code>/usr/local/lib/gal2-node/</code>	Installed SDK, demo, guard, uninstaller, manifests, and documentation, root-owned, mode <code>0755</code>
<code>/etc/systemd/system/gal2-node.service</code>	Rendered systemd unit, root-owned, mode <code>0644</code>
systemd enablement link	Created by <code>systemctl enable gal2-node.service</code>
Docker image	Exact frozen image tagged <code>gal2-node:rc11-amd64-evaluator-c8515d4</code>
Docker container	Named <code>gal2-node</code> while the service is running; launched with <code>--rm</code>

The installer enables `gal2-node.service` but does not start it automatically. The evaluator must configure the credential and explicitly start the service.

If the exact GAL-2 service is already active, the installer stops it before replacing the unit. A clean disposable evaluation host is recommended to avoid pre-existing state or name collisions.

9.1 SELinux behavior

The systemd unit uses Docker private relabeling on dedicated GAL-2 paths:

```
/run/gal2:/run/gal2:Z
/var/lib/gal2:/var/lib/gal2:Z
```

On an SELinux-enforcing host, `:z` can change the SELinux labels on those two dedicated host directories so that the private container can access them. The installer does not disable SELinux. Native RHEL 10.2 behavior, including host-side Provider access to the SHM file after relabeling, is part of the requested independent compatibility evaluation.

9.2 Actions the package does not perform

- It does not set the host time.
- It does not call a host-clock mutation primitive.
- It does not configure or restart `chronyd`, `NTP`, `PTP`, `GNSS`, or `systemd-timesyncd`.
- It does not use host networking.

- It does not mount the Docker socket inside the container.
- It does not run the container with `--privileged`.
- It drops all Linux capabilities.
- It enables `no-new-privileges`.
- It does not expose port 9095 beyond host loopback.
- It does not automatically start the Node before the API credential is configured.

10. Deployment architecture

The systemd unit requires `docker.service` and uses this security and runtime model:

```
systemd
-> Docker Engine
-> bridge-networked gal2-node container
-> HTTPS synchronization with api-v2.gal-2.com
-> local loopback HTTP contract
-> bind-mounted SHM and persistent state
```

Important Docker flags:

```
--rm
--pull never
--network bridge
--cap-drop ALL
--security-opt no-new-privileges:true
--publish 127.0.0.1:9095:9095
--volume /run/gal2:/run/gal2:Z
--volume /var/lib/gal2:/var/lib/gal2:Z
```

The prestart and stop guard validates the container name, image object, expected tag, and GAL-2 ownership labels before taking action. Unexpected objects cause refusal rather than deletion or replacement.

11. RHEL 10.2 x86_64 evaluation environment

11.1 Docker support boundary

Red Hat does not ship or support Docker Engine as the RHEL container engine. RHEL uses Podman. Docker's own documentation currently lists maintained RHEL 10 as an installation target for Docker Engine.

References:

- [Docker Engine installation on RHEL](#)
- [Red Hat RHEL 10 container-engine boundary](#)

RC11 requires genuine Docker Engine 28.3.3 or newer and an active `docker.service`. Do not substitute Podman or `podman-docker`.

Recommended first target:

- Disposable RHEL 10.2 x86_64 VM
- VM hosted on native x86_64 hardware
- Snapshot taken before Docker or GAL-2 installation
- SELinux left in its normal enforcing state
- Docker Engine installed using Docker's documented RHEL procedure
- No production workloads on the host

A VM on native x86_64 hardware is sufficient for initial installation, SELinux, systemd, Docker, SHM, contract, Provider, and fault-injection compatibility testing. Bare-metal x86_64 is preferred for later timing, latency, scheduling, and endurance characterization.

Docker installation itself can modify host networking, forwarding, firewall rules, packages, and services. Those are Docker platform changes, not changes performed by the GAL-2 installer. Use a disposable host and record the Docker configuration before installing GAL-2.

11.2 Preflight capture

Run before GAL-2 installation:

```
uname -m
cat /etc/redhat-release
uname -r
getenforce

systemctl is-active docker
systemctl is-enabled docker
sudo docker version
sudo docker info
sudo docker network inspect bridge >/dev/null && echo DOCKER_BRIDGE=PASS

command -v python3
command -v curl
command -v sha256sum
command -v tar
command -v unzip
command -v gpg
```

For the complete fault-injection harness, also record:

```
command -v iptables
sudo iptables -w 5 -nL DOCKER-USER
systemctl is-active firewalld || true
sudo firewall-cmd --state || true
```

Expected architecture:

x86_64

If `iptables -nL DOCKER-USER` fails, report the Docker firewall backend, `firewalld` state, command output, and host configuration before proceeding to fault injection. Do not disable `firewalld`, change the backend, or add an unreviewed compatibility rule merely to force the evaluation to pass.

Docker supports `iptables` and `nftables` firewall backends. The supplied RC11 fault helper specifically requires the `iptables` command and Docker-managed `DOCKER-USER` chain. A Docker native `nftables` configuration without that chain is outside this helper's supported fault-injection path.

References:

- [Docker packet filtering and firewalls](#)
- [Docker with nftables](#)

12. Exact deployment and baseline sequence

12.1 Review the plan before host mutation

From the extracted package directory:

```
python3 evaluation/run_external_evaluation.py --plan-only
```

Required final line:

```
GAL2_EXTERNAL_EVALUATION_PLAN=PASS
```

12.2 Install without starting

```
sudo ./scripts/install.sh
```

The installer validates:

- Tested payload checksums
- Frozen manifest
- Runtime archive identity
- OCI graph and all blob digests
- OCI labels and Linux AMD64 platform
- Loaded Docker object identity
- Installed ownership and modes

Required post-install state on a clean host:

```
gal2-node.service enabled
gal2-node.service inactive
GAL2_API_KEY empty until separately configured
```

12.3 Configure the credential

```
sudoedit /etc/gal2/node.env
```

Set the separately supplied credential:

```
GAL2_API_KEY=YOUR_GAL2_API_KEY
```

Never place the key in a repository, command transcript, issue, screenshot, evidence archive, or evaluation report.

Verify metadata without printing the credential:

```
sudo stat -c 'OWNER=%U:G%MODE=%aPATH=%n' /etc/gal2/node.env
```

Required result:

```
OWNER=root:root MODE=600 PATH=/etc/gal2/node.env
```

12.4 Start and establish baseline

```
sudo gal2-node start
sleep 40
sudo gal2-node doctor
gal2-node contract
```

A consumption-ready baseline must include:

```
shm_publication=PASS
gal2_time_ready=PASS
DOCTOR=PASS
mode=LIVE
safe_to_consume=true
reason=fresh_api_sync
```

Run the supplied enrolled-application Provider demo:

```
PYTHONPATH=/usr/local/lib/gal2-node/sdk/python \
python3 /usr/local/lib/gal2-node/examples/python/gal2_demo.py
```

Required application-level properties:

```
status=ok
safe_to_consume=true
raw_host_time_fallback=false
```

On SELinux-enforcing RHEL, record whether the host-side Provider can read:

```
/run/gal2/contract-v1.shm
```

Also capture:

```
ls -lZ /run/gal2 /run/gal2/contract-v1.shm /var/lib/gal2
sudo docker inspect gal2-node --format \
  'status={{.State.Status}}image={{.Config.Image}}network={{.HostConfig.NetworkMode}}global_ipv6={{range .NetworkSettings
    .Networks}}{{.GlobalIPv6Address}}{{end}}'
```

13. Controlled external evaluation

Run from the extracted package directory with a new evidence path that does not already exist:

```
sudo python3 evaluation/run_external_evaluation.py \
  --execute \
  --evidence-dir /tmp/gal2-rc11-external-evidence \
  --release-archive ../GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz \
  --expected-release-sha256 d452a276af1bf4bd2abb11dfa01f14f22f7c72f36eda9a3019c4269ef36ef553 \
  --release-signature ../GAL2_NODE_LINUX_AMD64_1.0.0-rc11-evaluator.tar.gz.asc \
  --release-public-key ../GAL2_RELEASE_SIGNING_PUBLIC_KEY_RC11.asc
```

13.1 Policy used by the accelerated harness

Policy	Normal default	Accelerated evaluation
Holdover soft limit	21600 seconds, 6 hours	60 seconds
Holdover hard limit	259200 seconds, 72 hours	90 seconds
Rejoin guard	10 seconds	10 seconds
LIVE validity	45 seconds	45 seconds
Provider unsafe propagation timeout	Not applicable	10 seconds

The accelerated limits make state transitions testable in minutes. They do not claim equivalence to a real-duration 72-hour native AMD64 endurance run. The harness must restore the normal policy afterward.

13.2 Fault-injection mechanism

The helper:

- Requires root, iptables, Docker bridge networking, and DOCKER-USER.
- Resolves the current evaluator container IPv4 address.
- Refuses non-bridge networking.
- Refuses a container with a global IPv6 address because IPv4-only isolation would be incomplete.
- Creates the dedicated chain GAL2_EVALUATOR_API_BLOCK.
- Inserts a source-address rule at the beginning of DOCKER-USER.
- Rejects all forwarded IPv4 egress from that evaluator container.
- Stores managed state under /run/gal2-evaluator-firewall/ with restricted ownership and mode.
- Removes its rule, dedicated chain, and state during cleanup.
- Rolls back partially applied state on errors and handled signals.

This mechanism isolates the entire evaluator container's forwarded IPv4 egress. It does not alter the host clock or timing service configuration.

13.3 Expected behavioral sequence

```
LIVE
-> HOLDOVER
-> DEGRADED while policy still permits consumption
-> recovery through REJOIN or direct LIVE when within threshold
-> LIVE
-> HOLDOVER
-> DEGRADED
-> FAIL_CLOSED
-> restore default policy and upstream access
-> REJOIN or direct LIVE
-> LIVE
```

The harness permits direct LIVE recovery only when the observed delta is within the declared slew threshold. Otherwise, controlled REJOIN is expected.

13.4 Required final markers

```
HOST_TIMING_STATE_UNCHANGED=PASS
ORIGINAL_NODE_ENV_RESTORED=PASS
RESTORED_DEFAULT_LIVE=PASS
GAL2_SERVICE_STATE_RESTORED=PASS
GAL2_EXTERNAL_EVALUATION=PASS
```

14. Evaluation result classification and failure criteria

Report raw command output, relevant logs, contract records, and the generated evidence directory for every failure or blocker.

14.1 Supply-chain failure

Any of the following is a failure:

- Published checksum mismatch
- Unexpected GPG primary fingerprint
- Invalid detached signature
- Tested-payload mismatch
- OCI blob, manifest, config, platform, or label mismatch
- Loaded Docker object outside the exact two-digest identity set
- Installed file hash, size, mode, or ownership mismatch

Do not continue execution after a supply-chain identity failure.

14.2 Environment compatibility blocker

Classify these as environment or compatibility blockers, not as successful fault-injection results:

- Non-AMD64 host
- Podman or `podman-docker` substituted for Docker Engine
- Docker Engine below 28.3.3
- Inactive or unavailable `docker.service`
- Missing Docker bridge network
- Missing `iptables` command for the full harness
- Missing Docker-managed `DOCKER-USER` chain
- Docker native `nftables` configuration without the required chain
- Global IPv6 enabled on the evaluator container
- SELinux denial preventing the dedicated bind mounts or host Provider from reading SHM

Preserve the original environment and report the blocker before changing firewall or SELinux policy.

14.3 Functional failure

Examples include:

- Node fails to reach consumption-ready `LIVE` with a valid credential and reachable API.
- `gal2-node doctor` fails after the documented readiness interval.
- SHM publication does not progress.
- The Provider cannot consume an authorized `LIVE` or safe `DEGRADED` contract.
- Recovery does not reach the state accepted by the harness within its built-in bounded deadline.

14.4 Safety failure

Any of the following is a safety failure:

- The Provider returns a GAL-2 timestamp in `FAIL_CLOSED`.
- The Provider does not raise `ProviderUnsafeError` after unsafe state has propagated within the harness timeout.
- A `FAIL_CLOSED` Provider result contains `gal2_epoch_ns`.
- A Provider value moves backward across the tested recovery boundary.
- Contract state permits consumption after the declared policy no longer authorizes it.
- The application silently falls back to raw host time after Provider refusal.

Record observed `uncertainty_ms` and compare it with the contract's declared policy fields, validity, mode, reason, and `safe_to_consume` decision. Any internal inconsistency should be reported. Evaluator-defined uncertainty thresholds are welcome when they are labeled separately from the RC11 built-in acceptance criteria.

14.5 Host-boundary or cleanup failure

Any of the following is a failure:

- Host timing-service state changes across the evaluation.
- Original `/etc/gal2/node.env` bytes, owner, group, or mode are not restored.
- Managed firewall rules or state remain after cleanup.
- Default 6-hour and 72-hour policy values are not restored.
- Original GAL-2 service active and enabled state is not restored.
- Final default LIVE recovery fails while the API is available.

The supplied harness uses its own bounded transition deadlines. RC11 does not declare a separate maximum `FAIL_CLOSED` overshoot. Tighter timing thresholds are welcome when recorded as evaluator-defined tests with explicit acceptance criteria, independently of the supplied harness result.

15. Uninstall and complete removal

15.1 Normal uninstall

```
sudo /usr/local/lib/gal2-node/uninstall.sh
```

Normal uninstall verifies ownership and identity, then removes the service, container, control command, and ordinary installed support payload. It deliberately preserves:

```
/etc/gal2/node.env
/var/lib/gal2
/run/gal2
gal2-node:rc11-amd64-evaluator-c8515d4
/usr/local/lib/gal2-node/uninstall.sh
/usr/local/lib/gal2-node/FROZEN_TESTED_PAYLOAD.json
```

Because normal uninstall preserves `/etc/gal2/node.env`, it can preserve the API credential. Use complete purge when credential removal is required.

15.2 Complete purge

```
sudo /usr/local/lib/gal2-node/uninstall.sh --purge
```

Complete purge verifies target ownership and exact image identity, then removes the configuration, credential, persistent state, runtime SHM, exact verified evaluator image, purge helper, and frozen identity anchor.

The uninstaller requires the Docker daemon so it can establish container and image ownership. It refuses destructive removal if the expected installation or image identity cannot be established.

The downloaded ZIP, extracted package, and evaluator-created evidence directory remain outside the installed product paths. The evaluator may archive or remove them separately after preserving required evidence.

The temporary API credential should also be revoked or disabled by GAL-2 after the agreed evaluation window.

16. Bundled founder-operated evidence

The bundled external evidence records a PASS on:

```
Ubuntu 24.04.4 LTS
Linux x86_64 under QEMU on Apple Silicon
Docker Engine 29.7.2
```

Observed modes:

```
DEGRADED
FAIL_CLOSED
HOLDOVER
LIVE
REJOIN
```

Recorded validations include:

- Default LIVE Provider acceptance

- Safe DEGRADED Provider acceptance
- Recovery Provider acceptance
- Typed `ProviderUnsafeError` refusal in `FAIL_CLOSED`
- No Provider timestamp returned in `FAIL_CLOSED`
- Host timing state unchanged
- Original `node.env` restored byte-exact with root ownership and mode `0600`
- Original service state restored
- Managed firewall state removed
- Final default LIVE restoration

This evidence is founder-operated. It does not claim native AMD64 timing characterization, real-duration AMD64 endurance, RHEL 10.2 compatibility, independent validation, or production certification.

The bundled end-to-end run did not inject a new outage while a rejoin slew was already active. The relevant RC11 behavior is covered by unit-level regression and remains available for evaluator-defined additional testing.

17. Expected effort, credential window, and findings handling

The following planning estimates are non-contractual and begin only after a compatible Linux AMD64 host with genuine Docker Engine is available:

Activity	Planning estimate
Artifact verification, installation, configuration, and baseline checks	30 to 60 minutes
Supplied accelerated external harness and immediate evidence review	20 to 30 minutes
Evaluator-defined stress, endurance, or additional fault tests	At evaluator discretion

The founder-operated accelerated harness completed in approximately 12 minutes on the declared QEMU environment. Native RHEL behavior, dependency installation, security review, and investigation of blockers can materially change the elapsed time.

The temporary credential's activation and expiration dates are communicated privately in the corporate covering message, not committed to this public document. Before execution, the evaluator should confirm that the credential window is still active. GAL-2 will acknowledge, reproduce where possible, and classify returned findings through the agreed correspondence channel. No response-time SLA is asserted by this RC11 handoff.

18. Requested independent evaluator record

Please record and return:

- RHEL release and kernel
- Physical or virtual deployment and host CPU architecture
- Docker client and server versions
- Docker firewall backend if known
- `firewalld` state
- `iptables -nL DOCKER-USER` result
- SELinux mode and any AVC denials
- Installation result
- Loaded Docker object identity
- Container labels and network mode
- `gal2-node doctor` result
- Baseline Time Contract
- Python Provider demo result
- Host-side SHM readability and SELinux labels
- External harness result
- Complete generated evidence directory
- Cleanup and restoration result

- Any incompatibility, unexpected behavior, or documentation ambiguity
- Any evaluator-defined tests and their acceptance criteria, kept separate from the supplied RC11 harness criteria

Do not include the API credential in returned logs or evidence.

Send technical findings to support@gal-2.com or through the separately agreed evaluator correspondence channel.

19. Product boundary

The GAL-2 API delivers GAL-2 Time created by the Protected Core.

The GAL-2 Node makes GAL-2 Time locally consumable.

The Time Contract governs whether an enrolled application may consume it.

Keep your timing stack. Install the GAL-2 Node. Connect the application once.